MBG 1201 Introduction to Modern Biology

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Regulation of Gene Expression

- Prokaryotes and eukaryotes alter gene expression in response to their changing environment.

- In multicellular eukaryotes, gene expression regulates development and is responsible for differences in cell types.

- RNA molecules play many roles in regulating gene expression in eukaryotes.

Eukaryotic gene expression is regulated at many stages

- All organisms must regulate which genes are expressed at any given time.
- Both unicellular organisms and the cells of multicellular organisms continually turn genes on and off in response to signals from their external and internal environments.

- Regulation of gene expression is also essential for cell specialization in multicellular organisms, which are made up of different types of cells.

- To perform its own distinct role, each cell type must maintain a specific program of gene expression in which certain genes are expressed and others are not.

Differential Gene Expression

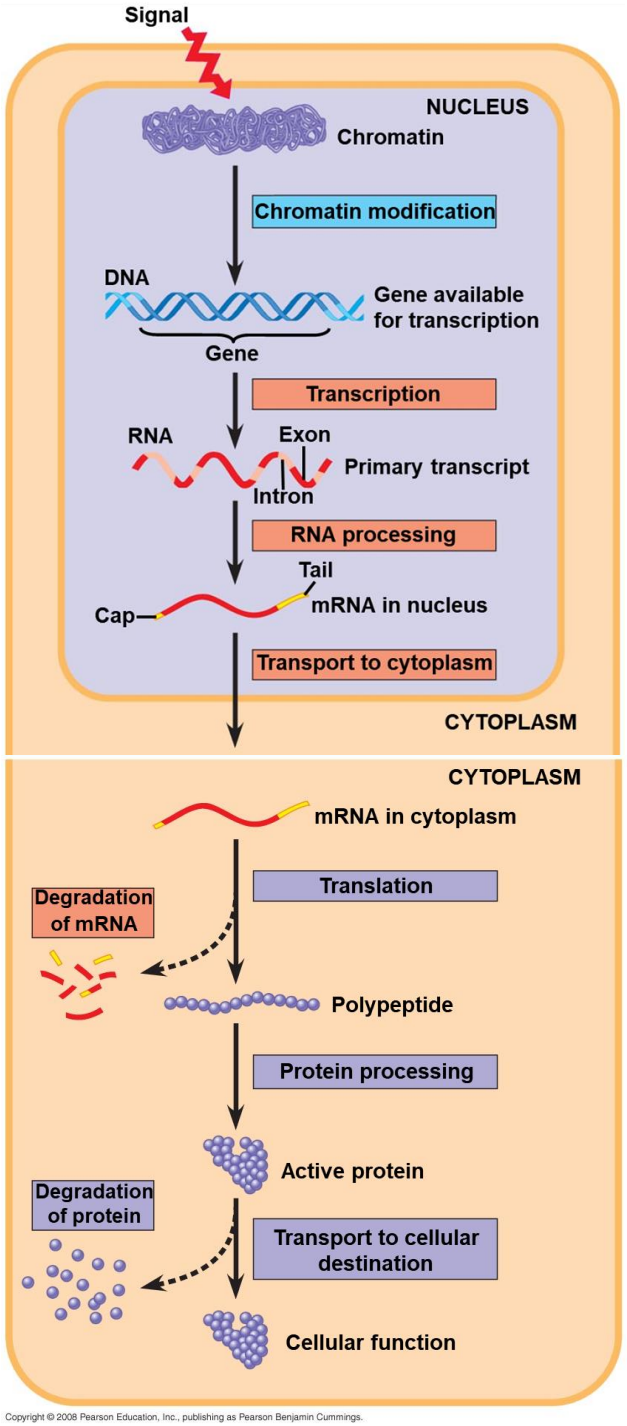
- Almost all the cells in an organism are genetically identical.
- Differences between cell types result from differential gene expression, the expression of different genes by cells with the same genome.
- Errors in gene expression can lead to diseases including cancer.
- Gene expression is regulated at many stages.

Eukaryotic gene expression is regulated at many stages

- A typical human cell might express about a third to a half of its protein-coding genes at any given time. Highly differentiated cells, such as muscle or nerve cells, express a smaller fraction of their genes. Almost all the cells in a multicellular organism contain an identical genome.

- A subset of genes is expressed in each cell type; some of these—about 35%—are "housekeeping" genes, expressed by many cell types, while others are unique to that cell type.
- The uniquely expressed genes allow these cells to carry out their specific function. The differences between cell types, therefore, are due not to different genes being present, but to differential gene expression, the expression of different genes by cells with the same genome.

Stages in gene expression that can be regulated in eukaryotic cells



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Each stage depicted in Figure is a potential control point at which gene expression can be **turned on or off, accelerated, or slowed down**. Many genes have more than one control point.

In this diagram, the colored boxes indicate the processes most often regulated; each color indicates the type of molecule that is affected (blue = DNA, red/orange = RNA, purple = protein). The nuclear envelope separating transcription from translation in eukaryotic cells allows for RNA processing, a form of post-transcriptional control that is not possible in prokaryotes. Eukaryotes also have a greater variety of control mechanisms operating before transcription and after translation. Some genes are regulated at multiple stages.

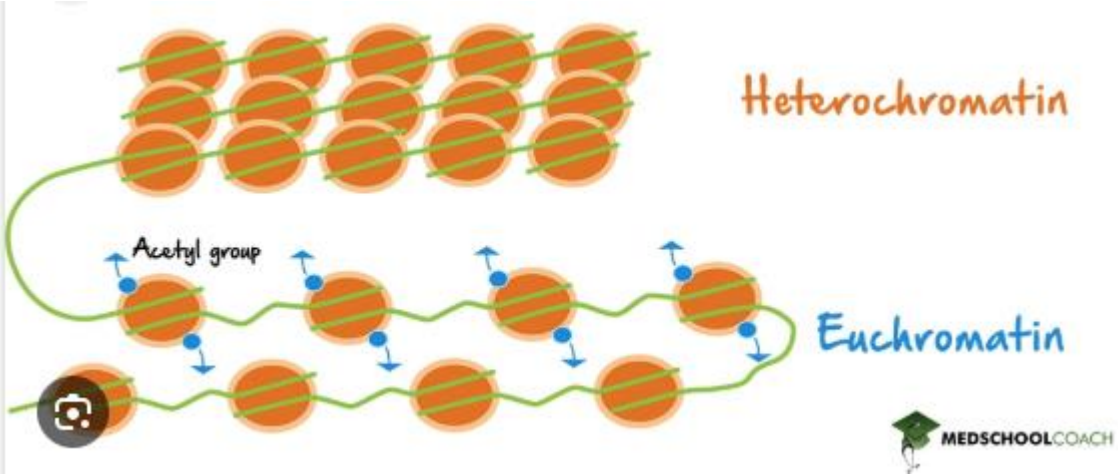
Regulation of Chromatin Structure

- Genes within highly packed heterochromatin are usually not expressed.

- Chemical modifications to histones and DNA of chromatin influence both chromatin structure and gene expression.

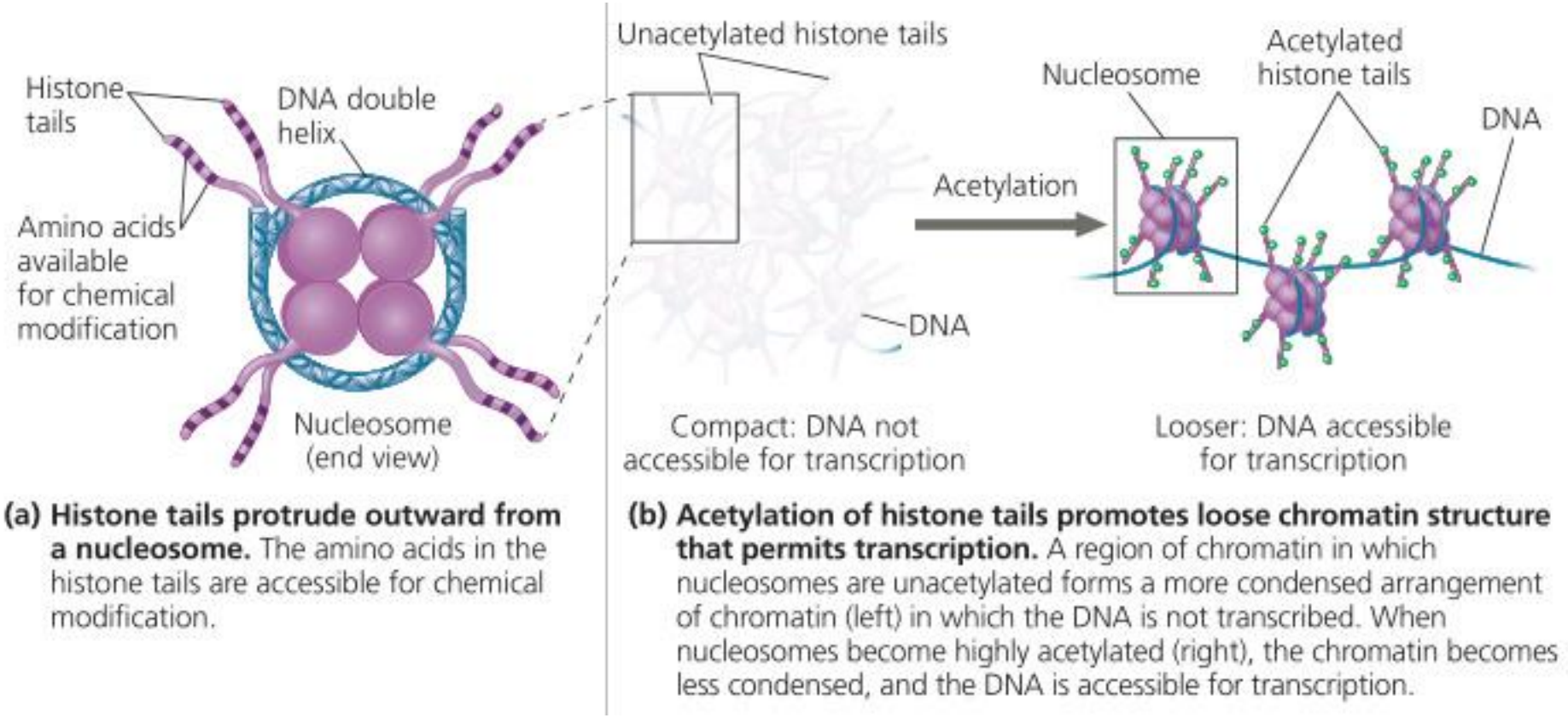
- Chromatin-modifying enzymes provide initial control of gene expression by making a region of DNA either more or less able to bind the transcription machinery.

- The N-terminus of each histone protein in a nucleosome protrudes outward from the nucleosome. These histone tails are susceptible to modification by enzymes that catalyze the addition or removal of specific chemical groups, such as acetyl, methyl, and phosphate groups.



Histone Modifications

- Generally, **histone acetylation**—the addition of an acetyl group to an amino acid in a histone tail—appears to promote transcription by opening up chromatin structure.
- **Histone methylation** -the addition of methyl groups to histones- can lead to the condensation of chromatin and reduced transcription.



(a) Histone tails protrude outward from a nucleosome. The amino acids in the histone tails are accessible for chemical modification.

(b) Acetylation of histone tails promotes loose chromatin structure that permits transcription. A region of chromatin in which nucleosomes are unacetylated forms a more condensed arrangement of chromatin (left) in which the DNA is not transcribed. When nucleosomes become highly acetylated (right), the chromatin becomes less condensed, and the DNA is accessible for transcription.

- Often, the addition of a particular chemical group may create a new binding site for enzymes that further modify chromatin structure.

DNA Methylation

- DNA methylation, the addition of methyl groups to certain bases in DNA, is associated with reduced transcription in some species. DNA methylation can cause long-term inactivation of genes in cellular differentiation.

- Once methylated, genes usually stay that way through successive cell divisions in a given individual. **At DNA sites where one strand is already methylated, enzymes methylate the correct daughter strand after each round of DNA replication.** Methylation patterns are thus passed on to daughter cells, and cells forming specialized tissues keep a chemical record of what occurred during embryonic development.
- A methylation pattern maintained in this way also accounts for **genomic imprinting** in mammals, where methylation permanently regulates expression of either the maternal or paternal allele of particular genes at the start of development.
- In genomic imprinting, methylation regulates expression of either the maternal or paternal alleles of certain genes at the start of development.

Epigenetic Inheritance

- The chromatin modifications do not change the DNA sequence, but they may be passed along to future generations of cells.

- Inheritance of traits transmitted by mechanisms not involving the nucleotide sequence is called **epigenetic inheritance**.

- **Epigenetics** is the study of epigenetic inheritance.

- Whereas mutations in the DNA are permanent changes, modifications to the chromatin can be reversed.

For example, DNA methylation patterns are largely erased during gamete formation and reestablished during embryonic development. Furthermore, they are changeable, thus responding more rapidly to environmental conditions.

Epigenetic Inheritance



(a) Effects of maternal diet on genetically identical mice. These two mice are genetically identical (*agouti* mutants), but their mothers were fed different diets. A maternal diet without compounds that donate methyl groups resulted in the obese, yellow offspring on the left. A diet supplemented with methyl-donating compounds, such as folic acid, led to the speckled brown fur (normal *agouti* phenotype) and normal weight of the mouse on the right.

A mouse mutant whose genome had been altered so that a gene called agouti that determines coat color, normally expressed only briefly during fur formation, was instead expressed throughout development.
This overexpression resulted in yellow mice,
left mouse) rather than the usual brownish (“agouti”) color.
Supplementing the diet of pregnant mothers with methyl group–containing compounds (like folic acid) could shift the range of coat colors of the offspring back to normal (Figure right mouse).
Feeding methyl groups to the mothers at a key time during gestation led to a change in gene expression in the offspring’s phenotype. Further studies showed that the effects were even observed in the next generation—the “grandpups” of the original female mouse.

Epigenetic Inheritance

- Epigenetic variations might help explain cases where one identical twin acquires a genetically based disease, such as schizophrenia, but the other does not, despite their identical genomes.

- Alterations in normal patterns of DNA methylation are seen in some cancers, where they are associated with inappropriate gene expression.

- Enzymes that modify chromatin structure by adding methyl or acetyl groups are integral parts of the eukaryotic cell's machinery for regulating transcription.

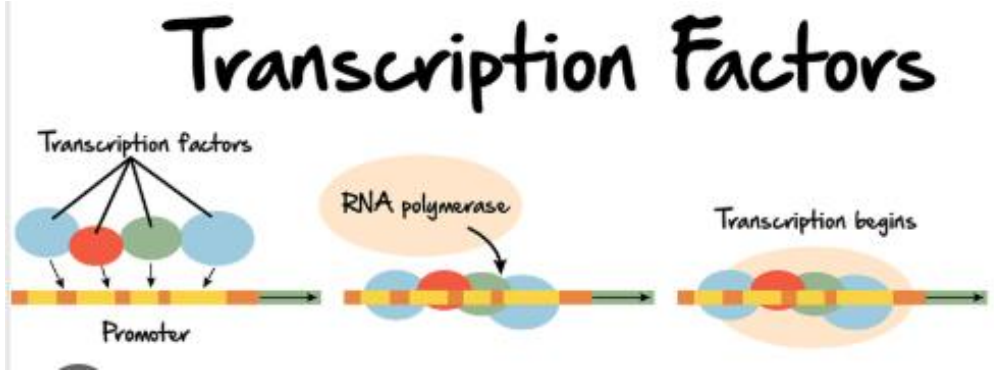
Regulation of Transcription Initiation

- Once the chromatin of a gene is optimally modified for expression, the initiation of transcription is the next major step at which gene expression is regulated.

- As in bacteria, the regulation of transcription initiation in eukaryotes involves proteins that bind to DNA and either facilitate or inhibit binding of RNA polymerase.

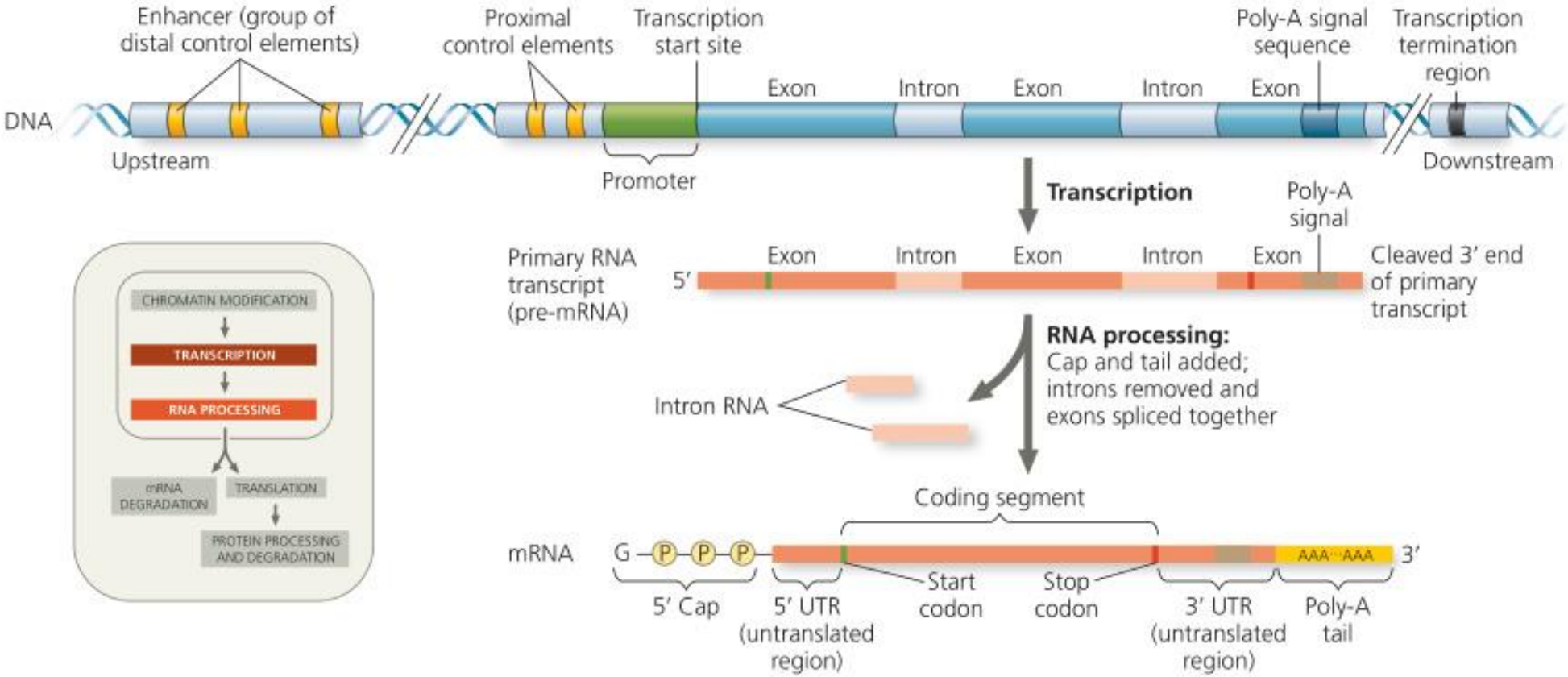
  - Associated with most eukaryotic genes are multiple **control elements**, segments of noncoding DNA that serve as binding sites for the proteins called **transcription factors**, which bind to the control elements and regulate transcription.

  - Control elements on the DNA and the transcription factors that bind to them are critical to the precise regulation of gene expression seen in different cell types.



Organization of a Typical Eukaryotic Gene

Each eukaryotic gene has a promoter—a DNA sequence where RNA polymerase II binds and starts transcription, proceeding "downstream." A number of control elements (gold) are involved in regulating the initiation of transcription; these are DNA sequences located near (proximal to) or far from (distal to) the promoter. Distal control elements can be grouped together as enhancers, one of which is shown for this gene.



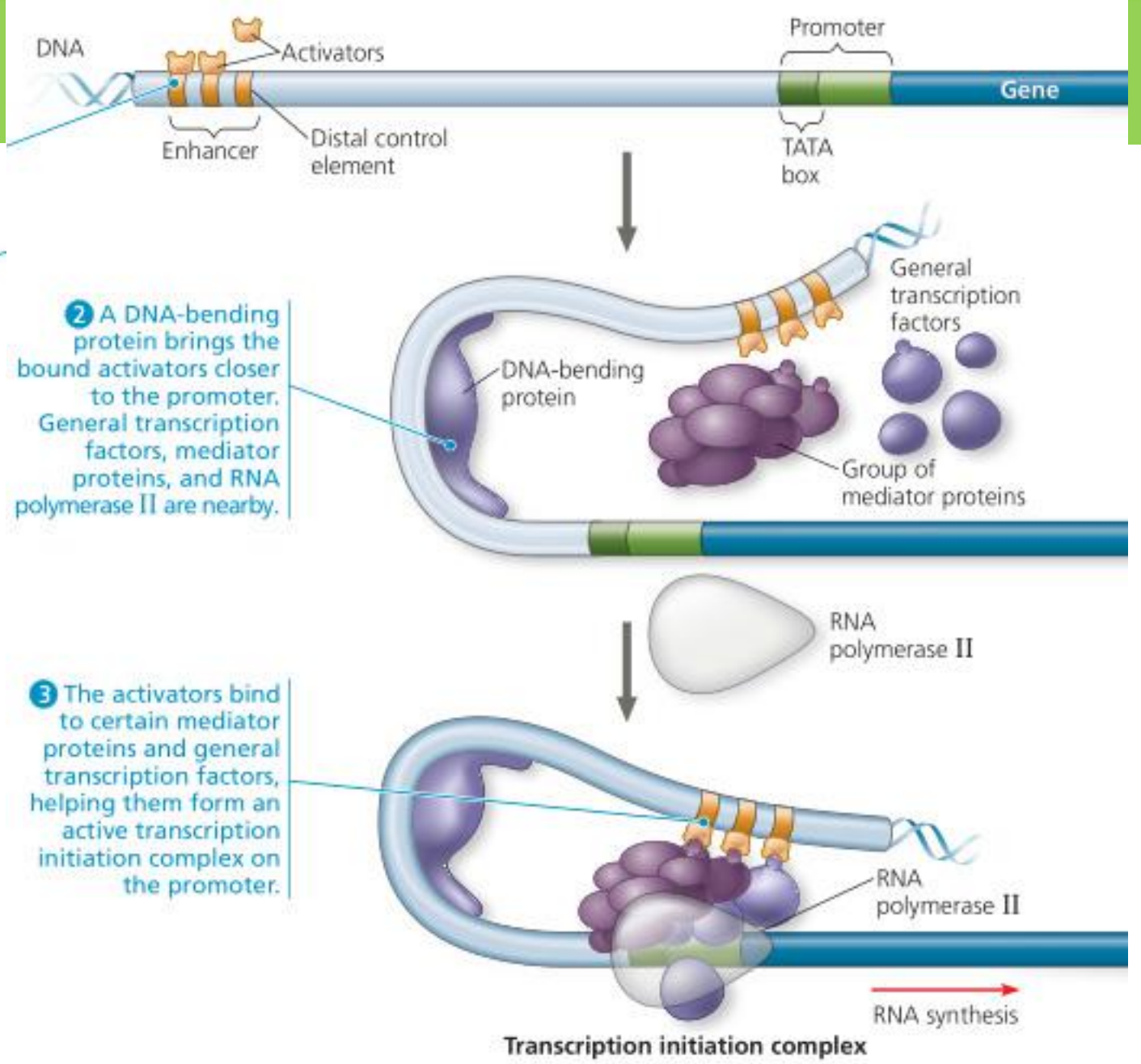
General Transcription Factors

- To initiate transcription, eukaryotic RNA polymerase requires the assistance of transcription factors.
- **General transcription** factors are essential for the transcription of all protein-coding genes.
- A few general transcription factors bind to a DNA sequence, such as the TATA box in most promoters, but many bind to proteins, including other transcription factors as well as RNA polymerase II.
- Protein-protein interactions are crucial to the initiation of eukaryotic transcription.
- Only when the complete initiation complex has assembled can the polymerase begin to move along the DNA template strand and transcribe.

Specific Transcription Factors

- Some genes are expressed all the time, but others are not; instead, they are regulated.
- For these genes, the interaction of general transcription factors and RNA polymerase II with a promoter usually leads to a low rate of initiation and production of few RNA transcripts from genes that are not expressed at significant levels all the time or in all cells.
- In eukaryotes, high levels of transcription of these particular genes at the appropriate time and place depend on the interaction of control elements with another set of proteins, which can be thought of as **specific transcription factors**.
- In eukaryotes, the rate of gene expression can be strongly increased or decreased by the binding of specific transcription factors, either **activators or repressors**, to the control elements of enhancers.

Enhancers and Activators (Specific Transcription Factors)



An **activator** is a protein that binds to an enhancer and stimulates transcription of a gene.

Protein-mediated bending of the DNA brings the bound activators into contact with a group of mediator proteins, which in turn interact with general transcription factors at the promoter. These protein-protein interactions help assemble and position the initiation complex on the promoter.

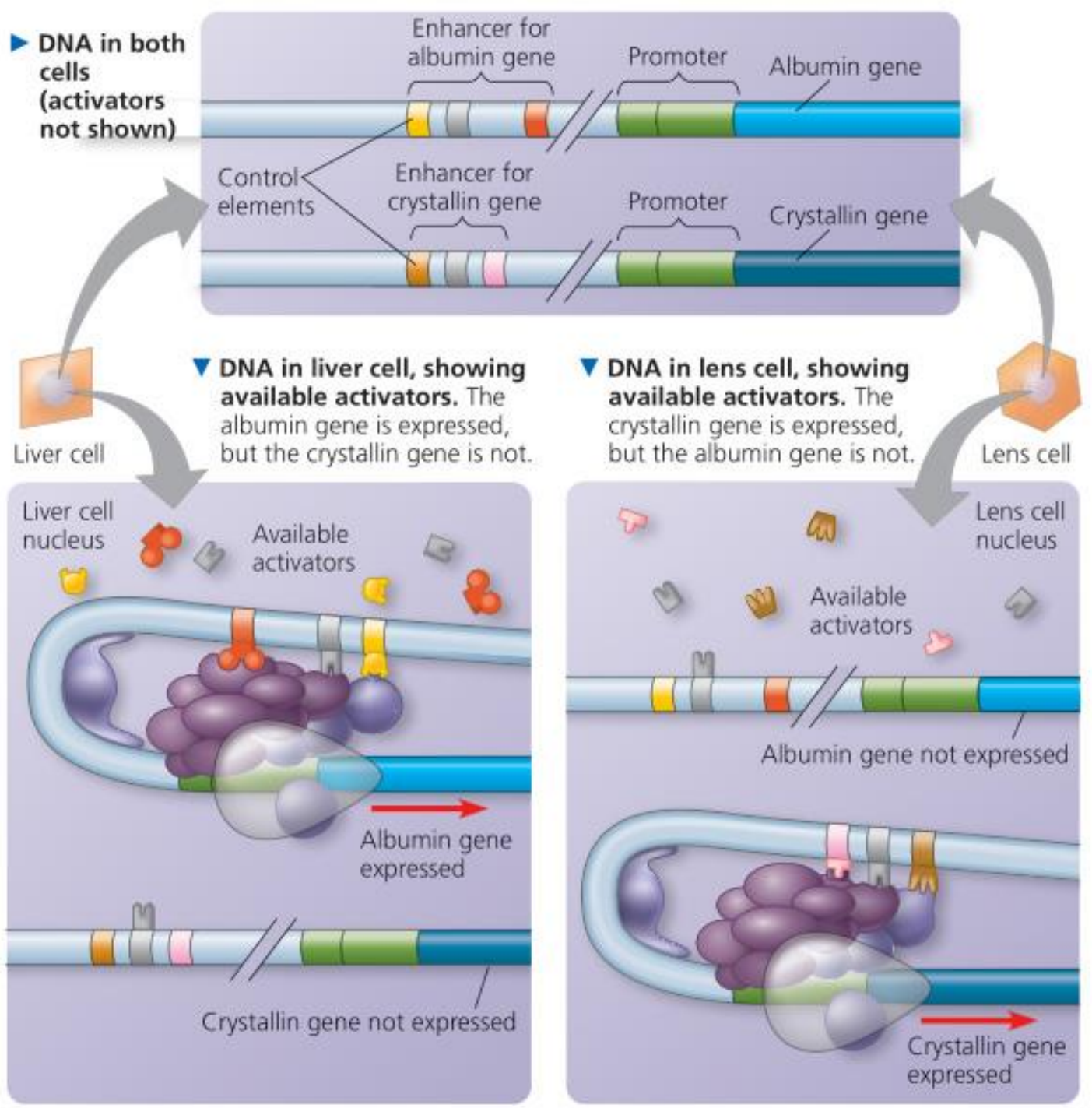
Enhancers and Repressors

- Some transcription factors function as **repressors**, inhibiting expression of a particular gene.
- Some repressors bind directly to control element DNA (in enhancers or elsewhere), blocking activator binding.
- Other repressors interfere with the activator itself so it can't bind the DNA.

- Some activators and repressors act indirectly by influencing chromatin structure to promote or silence transcription.
- For example some activators recruit proteins that acetylate histones near the promoters of specific genes, thus promoting transcription.
- Similarly, some repressors recruit proteins that remove acetyl groups from histones, leading to reduced transcription, a phenomenon referred to as **silencing**.

Combinatorial Control of Gene Activation

- A particular combination of control elements can activate transcription only when the appropriate activator proteins are present.
- The specific transcription factors made in a cell determine which genes are expressed.



Coordinately Controlled Genes in Eukaryotes

- How does the eukaryotic cell deal with a group of genes of related function that need to be turned on or off at the same time?

- Unlike the genes of a prokaryotic operon which is regulated by a single promoter and transcribed into a single mRNA molecule , each of the coordinately controlled eukaryotic genes has a promoter and control elements.

- These genes can be scattered over different chromosomes, but each has the same combination of control elements.

- Copies of the activators recognize specific control elements and promote simultaneous transcription of the genes.
- Genes with the same sets of control elements are activated by the same chemical signals.

Mechanisms of Post-Transcriptional Regulation

- Transcription alone does not account for gene expression.
- Many regulatory mechanisms operate at the various stages after transcription.
- These mechanisms allow a cell to rapidly finetune gene expression in response to environmental changes without altering its transcription patterns.

RNA Processing

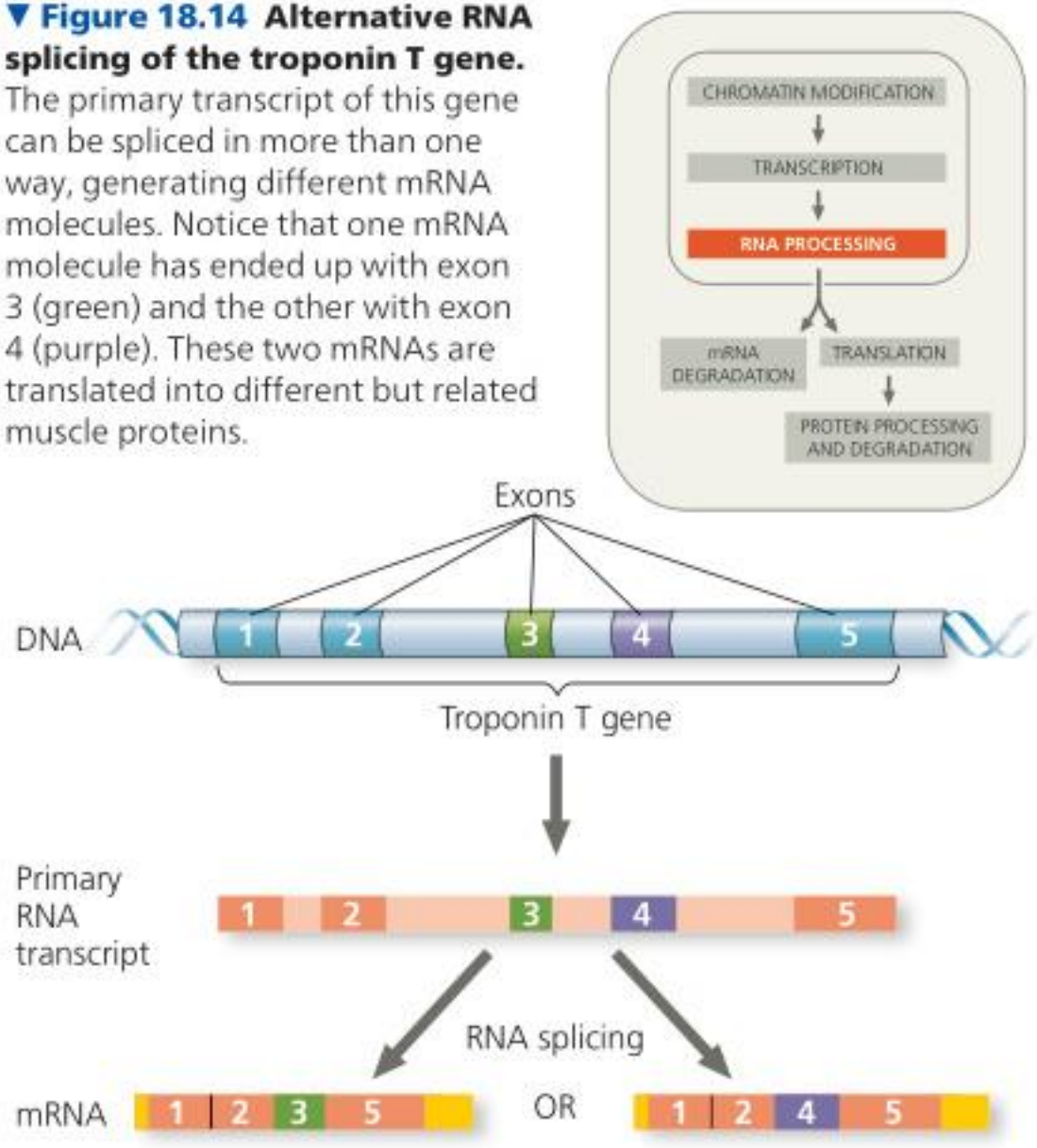
One example of regulation at the RNA-processing level is **alternative RNA splicing**.

In alternative RNA splicing, different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns.

A simple example of alternative RNA splicing is shown in Figure 18.14 for the troponin T gene, which encodes multiple closely related proteins with slightly different effects on muscle contraction.

More than 90% of human protein-coding genes likely undergo alternative splicing. Thus, the extent of alternative splicing greatly multiplies the number of possible human proteins, which may be better correlated with complexity of form than the number of genes.

▼ Figure 18.14 Alternative RNA splicing of the troponin T gene. The primary transcript of this gene can be spliced in more than one way, generating different mRNA molecules. Notice that one mRNA molecule has ended up with exon 3 (green) and the other with exon 4 (purple). These two mRNAs are translated into different but related muscle proteins.



mRNA Degradation

- The life span of mRNA molecules in the cytoplasm is important in determining the pattern of protein synthesis in a cell.
- Eukaryotic mRNA is more long lived than prokaryotic mRNA.
- Nucleotide sequences that affect how long an mRNA remains intact are often found in the untranslated region at the 3’ end of the molecule.
- In one experiment, researchers transferred such a sequence from the short-lived mRNA for a growth factor to the 3’ end of a normally stable globin mRNA. The globin mRNA was quickly degraded.

Initiation of Translation

- Translation is another stage where gene expression is regulated, most commonly at the initiation stage.

- For some mRNAs, the initiation of translation can be blocked by regulatory proteins that bind to specific sequences or structures within the untranslated region (UTR) at the 5′ or 3′ end, preventing the attachment of ribosomes.

- Alternatively, translation of all mRNAs in a cell may be regulated simultaneously.

- For example, translation initiation factors are simultaneously activated in an egg following fertilization.

Protein Processing and Degradation

- After translation, various types of protein processing, including cleavage and the addition of chemical groups, are subject to control.
- Eukaryotic polypeptides must be processed to yield functional protein molecules.
  e.g. Pro-insulin is cleaved to form functional insulin hormone.

- In addition, many proteins undergo chemical modifications that make them functional.
- Regulatory proteins are commonly activated or inactivated by the reversible addition of phosphate groups. Sugar molecules are attached to proteins destined for the surface of animal cells

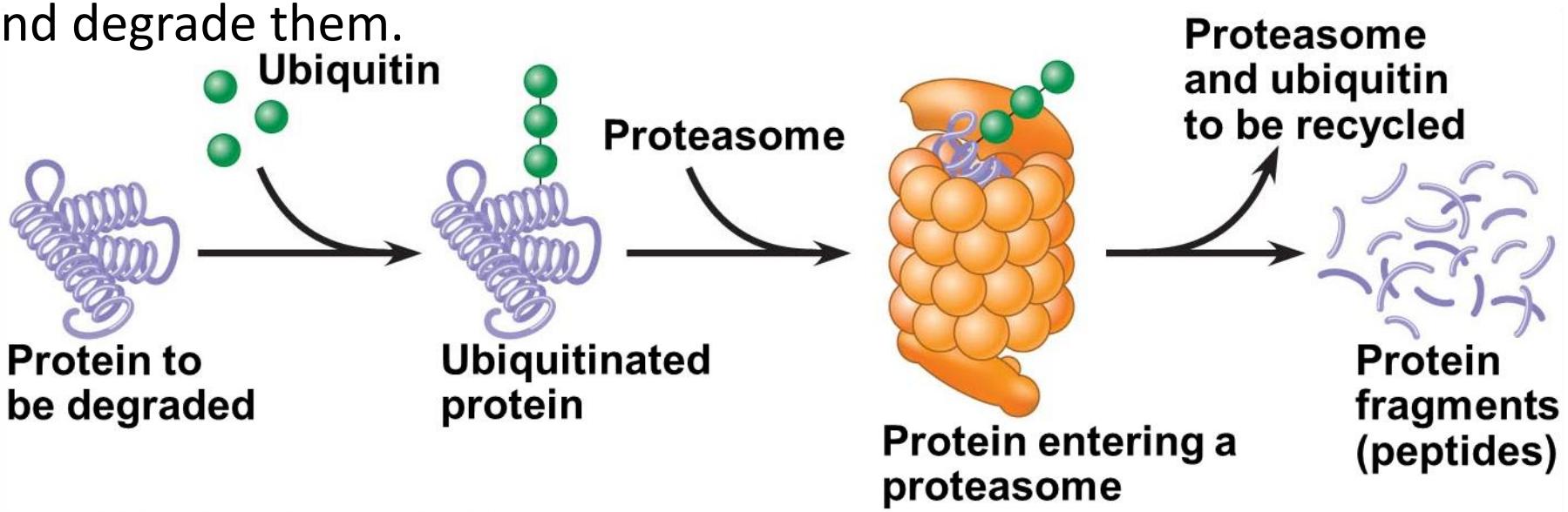
- Cell-surface proteins and many others must also be transported to target destinations in the cell in order to function.
- Regulation might occur at any of the steps involved in modifying or transporting a protein.

Protein Processing and Degradation

- Finally, the length of time each protein functions in the cell is strictly regulated by selective degradation.

e.g. the cyclins involved in regulating the cell cycle, must be relatively short-lived if the cell is to function appropriately.

- To mark a protein for destruction, the cell commonly attaches molecules of a small protein called ubiquitin to the protein.
- **Proteasomes** are giant protein complexes that bind ubiquitin-tagged protein molecules and degrade them.



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Noncoding RNAs play multiple roles in controlling gene expression

- Only a small fraction of DNA (approximately 1.5% of the human genome) codes for proteins, rRNA, and tRNA.
- A significant amount of the genome may be transcribed into noncoding RNAs.
- Noncoding RNAs regulate gene expression at two points: mRNA translation and chromatin configuration.

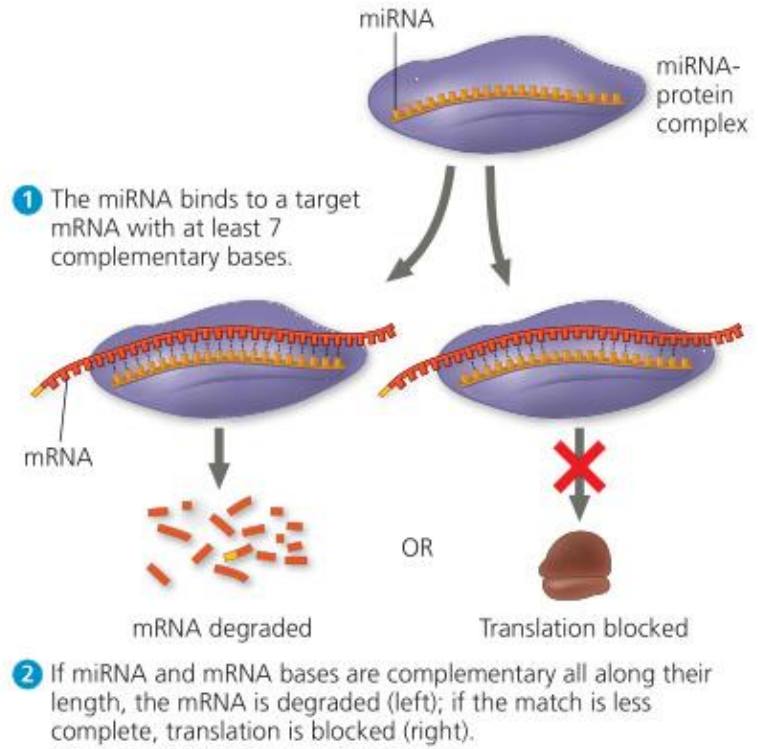
Effects on mRNAs by MicroRNAs and Small Interfering RNAs

- MicroRNAs (miRNAs) are small, single-stranded RNA moleculesof about 22 nucleotides that forms a complex with one or more proteins.
- A longer RNA precursor is processed by cellular enzymes into an miRNA.

- The miRNA allows the complex to bind to any mRNA molecule with at least seven or eight nucleotides of complementary sequence.
- The miRNA-protein complex then degrades the target mRNA or, less often, simply blocks its translation.

- Small interfering RNAs (siRNAs) are similar in size and function to miRNAs.

- **RNA interference** (RNAi) is the <u>blocking of gene expression by siRNAs.</u>



Chromatin Remodeling and Effects on Transcription by ncRNAs

- In addition to regulating mRNAs, small noncoding RNAs can cause remodeling of chromatin structure.

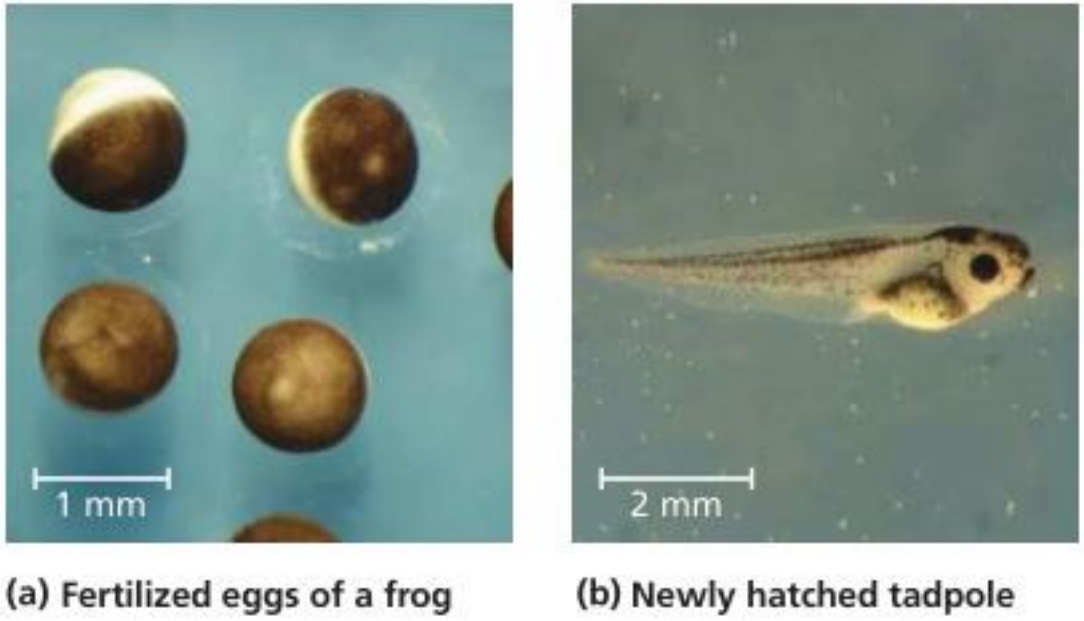
- siRNAs play a role in heterochromatin formation and can block large regions of the chromosome.

- Small RNAs may also block transcription of specific genes.

A Genetic Program for Embryonic Development

- During embryonic development, a fertilized egg gives rise to many different cell types.
- Gene expression orchestrates the developmental programs of animals.

▼ Figure 18.16 From fertilized egg to animal: What a difference four days makes. It takes just four days for cell division, differentiation, and morphogenesis to transform each of the fertilized frog eggs shown in **(a)** into a tadpole like the one in **(b)**.



(a) Fertilized eggs of a frog (b) Newly hatched tadpole

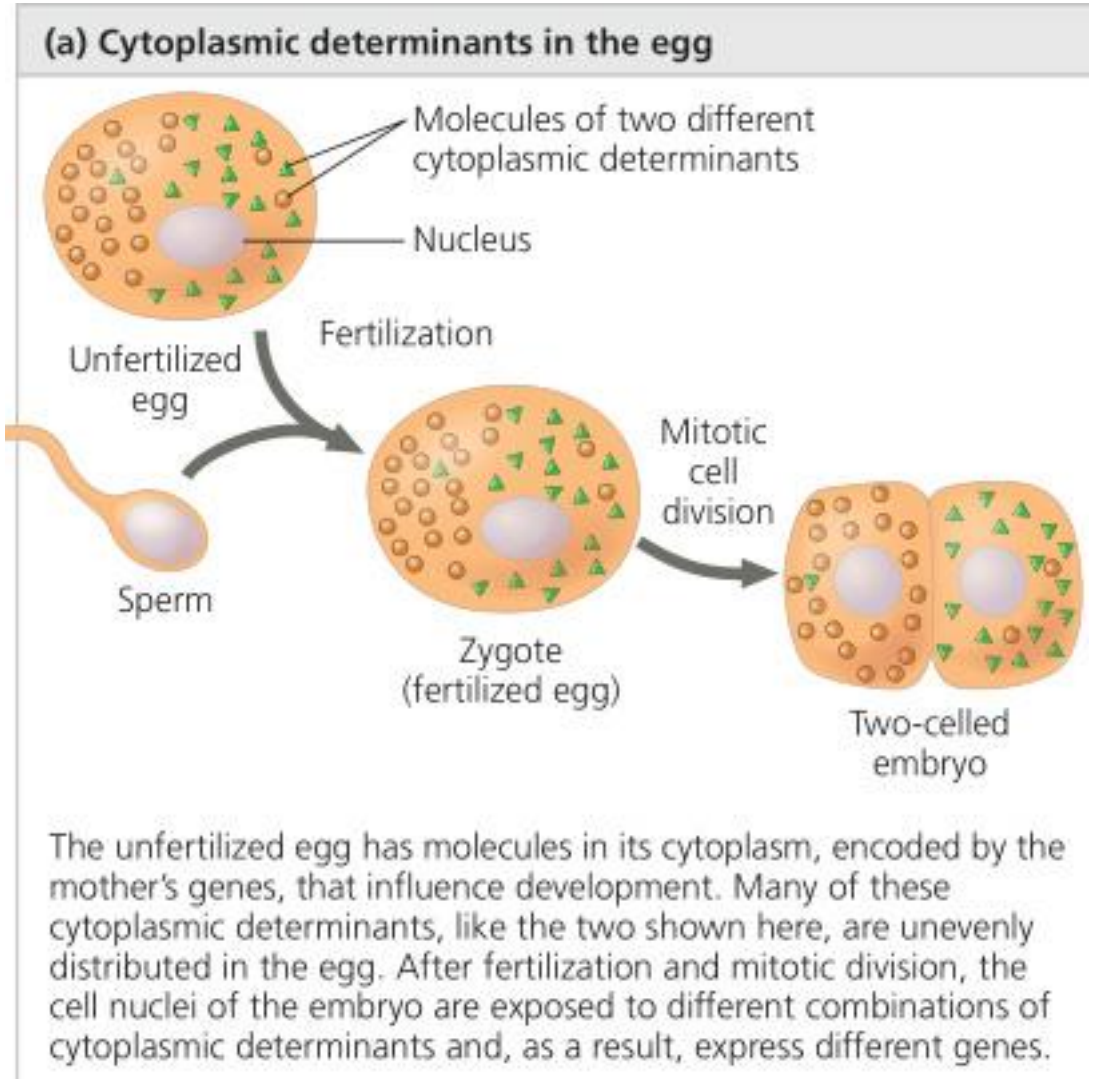
- The transformation from zygote to adult results from cell division, cell differentiation, and morphogenesis.
- Cell differentiation is the process by which cells become specialized and take on specific functions.
- Morphogenesis is the development of the form of an organism and its structures.
- Differential gene expression results from genes being regulated differently in each cell type.
- Materials in the egg can set up gene regulation that is carried out as cells divide.

Cytoplasmic Determinants and Inductive Signals

- An egg's cytoplasm contains RNA, proteins, and other substances that are distributed unevenly in the unfertilized egg.

- **Cytoplasmic determinants** are maternal substances in the egg that influence early development.

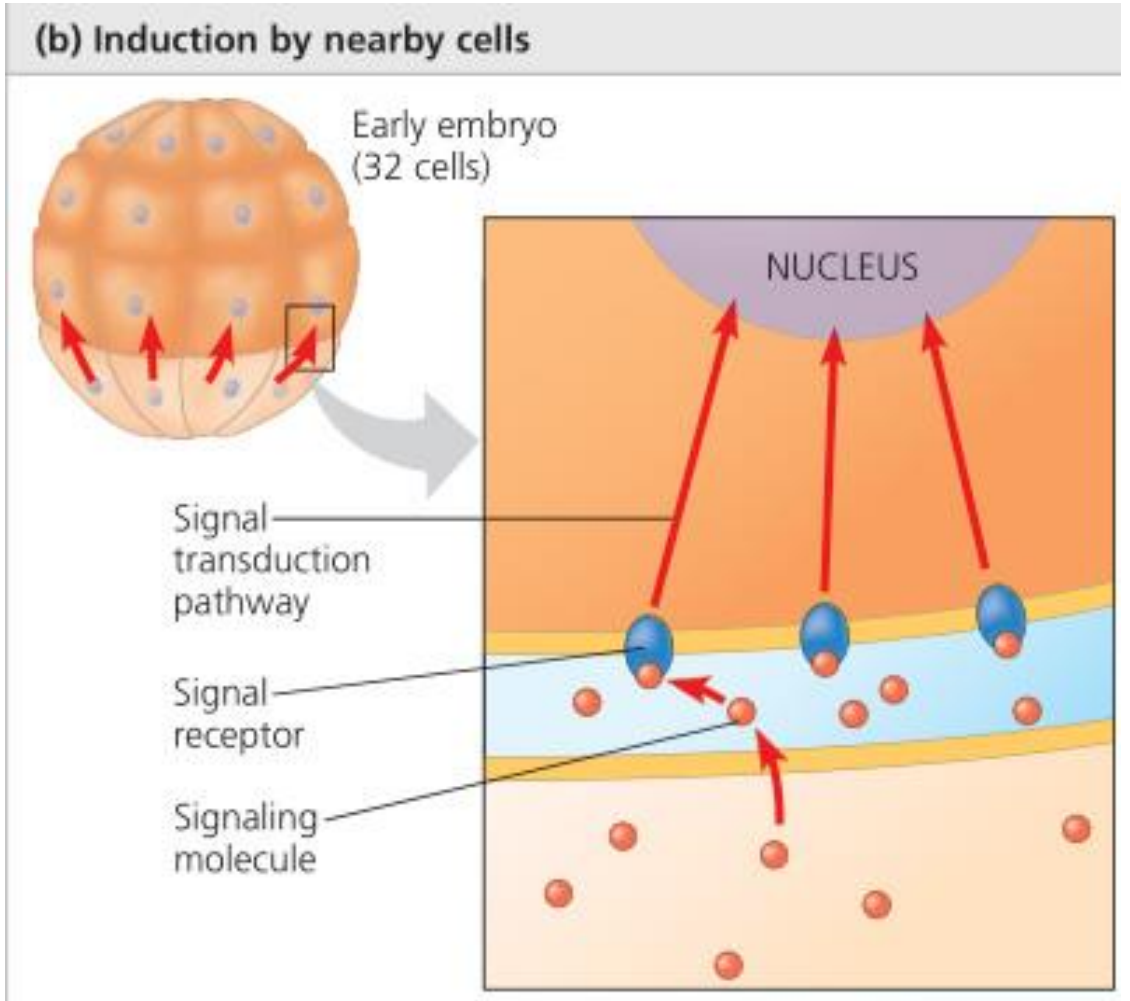
- As the zygote divides by mitosis, cells contain different cytoplasmic determinants, which lead to different gene expression.



The unfertilized egg has molecules in its cytoplasm, encoded by the mother's genes, that influence development. Many of these cytoplasmic determinants, like the two shown here, are unevenly distributed in the egg. After fertilization and mitotic division, the cell nuclei of the embryo are exposed to different combinations of cytoplasmic determinants and, as a result, express different genes.

Cytoplasmic Determinants and Inductive Signals

- The other important source of developmental information is the environment around the cell, especially signals from nearby embryonic cells.
- In the process called **induction**, signal molecules from embryonic cells cause transcriptional changes in nearby target cells.
- In general, the signal sends a cell down a specific developmental path by causing changes in its gene expression that lead to observable cellular changes.
- Thus, interactions between embryonic cells induce differentiation of specialized cell types.



Cells in the lowest tier of cells are releasing molecules that signal (induce) nearby cells to change their gene expression.

Sequential Regulation of Gene Expression During Cellular Differentiation

- **Determination** refers to the point at which an embryonic cell is irreversibly committed to becoming a particular cell type. Determination commits a cell to its final fate.

- Once it has undergone determination, an embryonic cell can be experimentally placed in another location in the embryo and it will still differentiate into the cell type that is its normal fate.

- **Differentiation**, then, is the process by which a cell attains its determined fate.

- Thus, determination precedes differentiation.

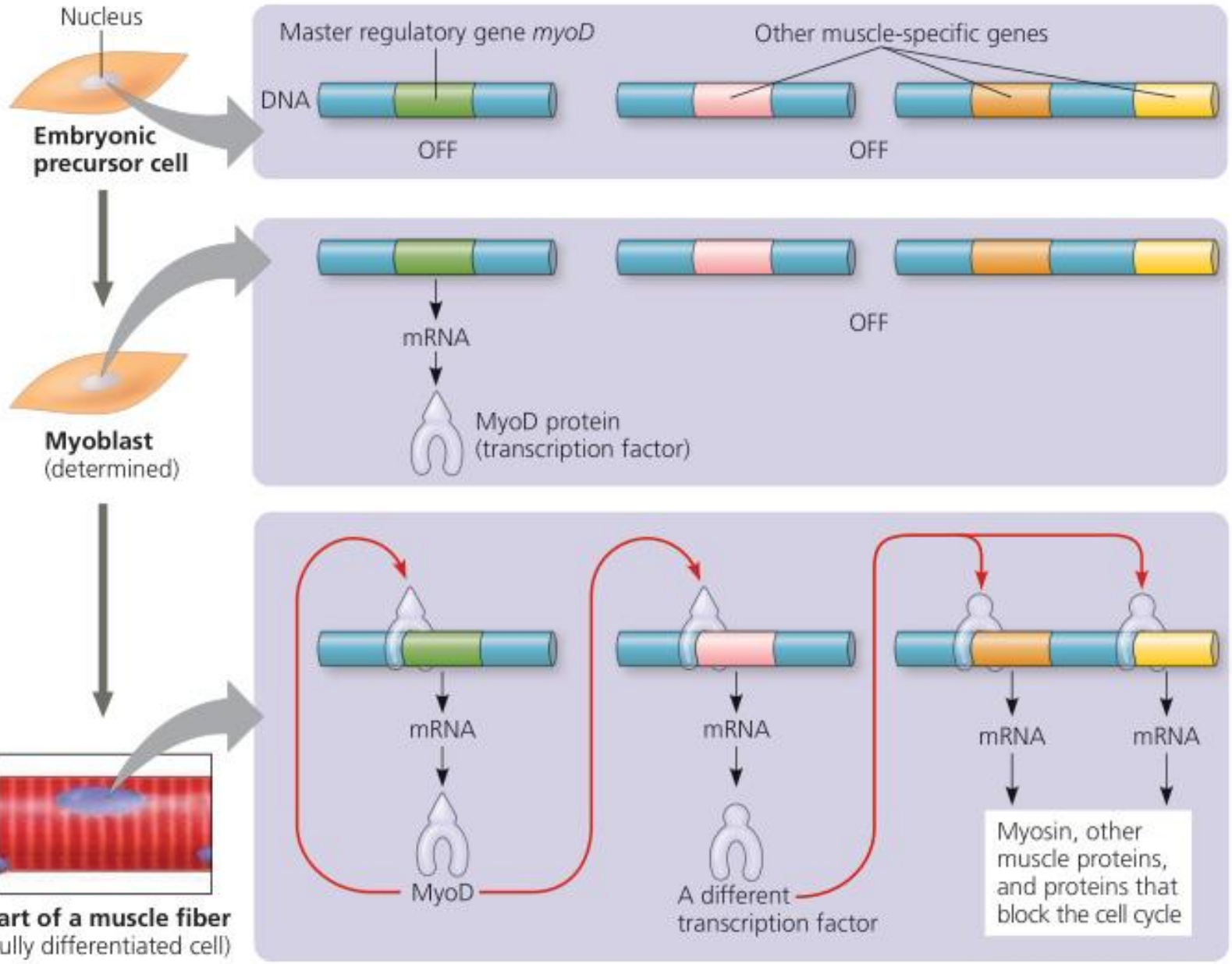
Sequential Regulation of Gene Expression During Cellular Differentiation

- Today we understand determination in terms of molecular changes that result in observable cell differentiation, marked by the expression of genes for tissue-specific proteins.
- Such proteins are found only in a specific cell type and give the cell its characteristic structure and function.
- The first sign of differentiation is the appearance of mRNAs for tissue-specific proteins.
- Differentiated cells are specialists at making tissue-specific proteins.
- For example, as a result of transcriptional regulation, liver cells specialize in making albumin, and lens cells specialize in making crystallin.

Muscle cell determination

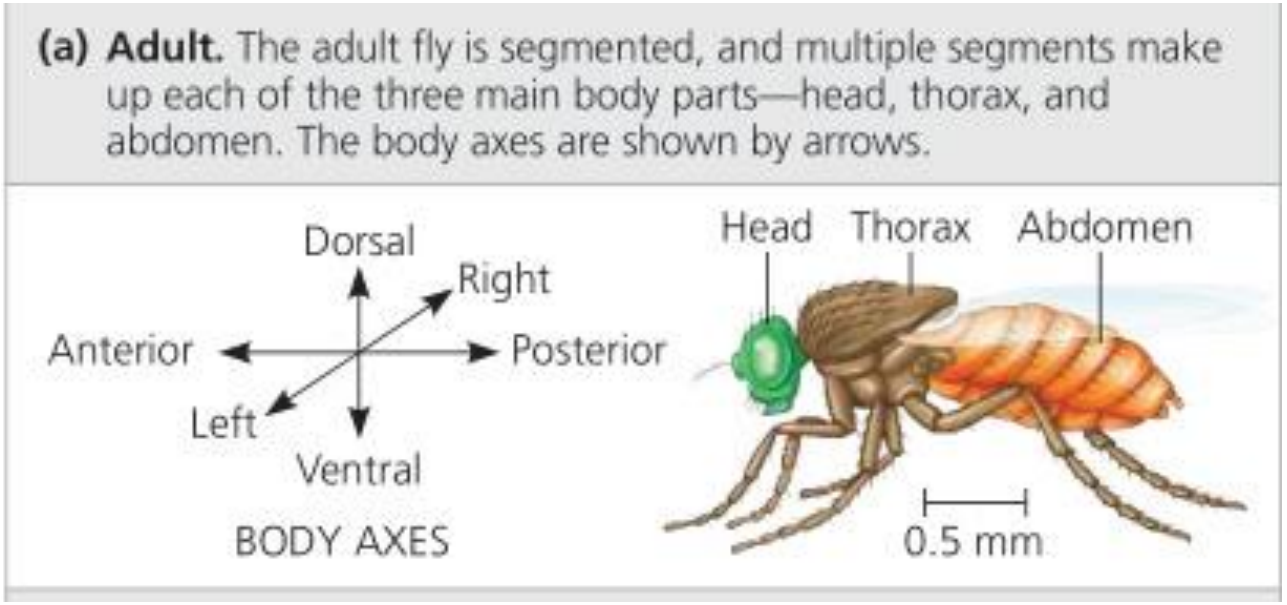
- Embryonic precursor cells have the potential to develop into a number of cell types.
- When determination occurs some of them develop into myoblasts.
- Myoblasts produce muscle-specific proteins and form skeletal muscle cells.
- MyoD is one of several "**master regulatory genes**" that produce proteins that commit the cell to becoming skeletal muscle.
- The MyoD protein is a transcription factor that binds to enhancers of various target genes.
- Thus, in the case of muscle cells, the molecular basis of determination is the expression of one or more of these master regulatory genes.

▼ **Figure 18.18 Determination and differentiation of muscle cells.** Skeletal muscle cells arise from embryonic cells as a result of changes in gene expression. (In this depiction, the process of gene activation is greatly simplified.)



Pattern Formation: Setting Up the Body Plan

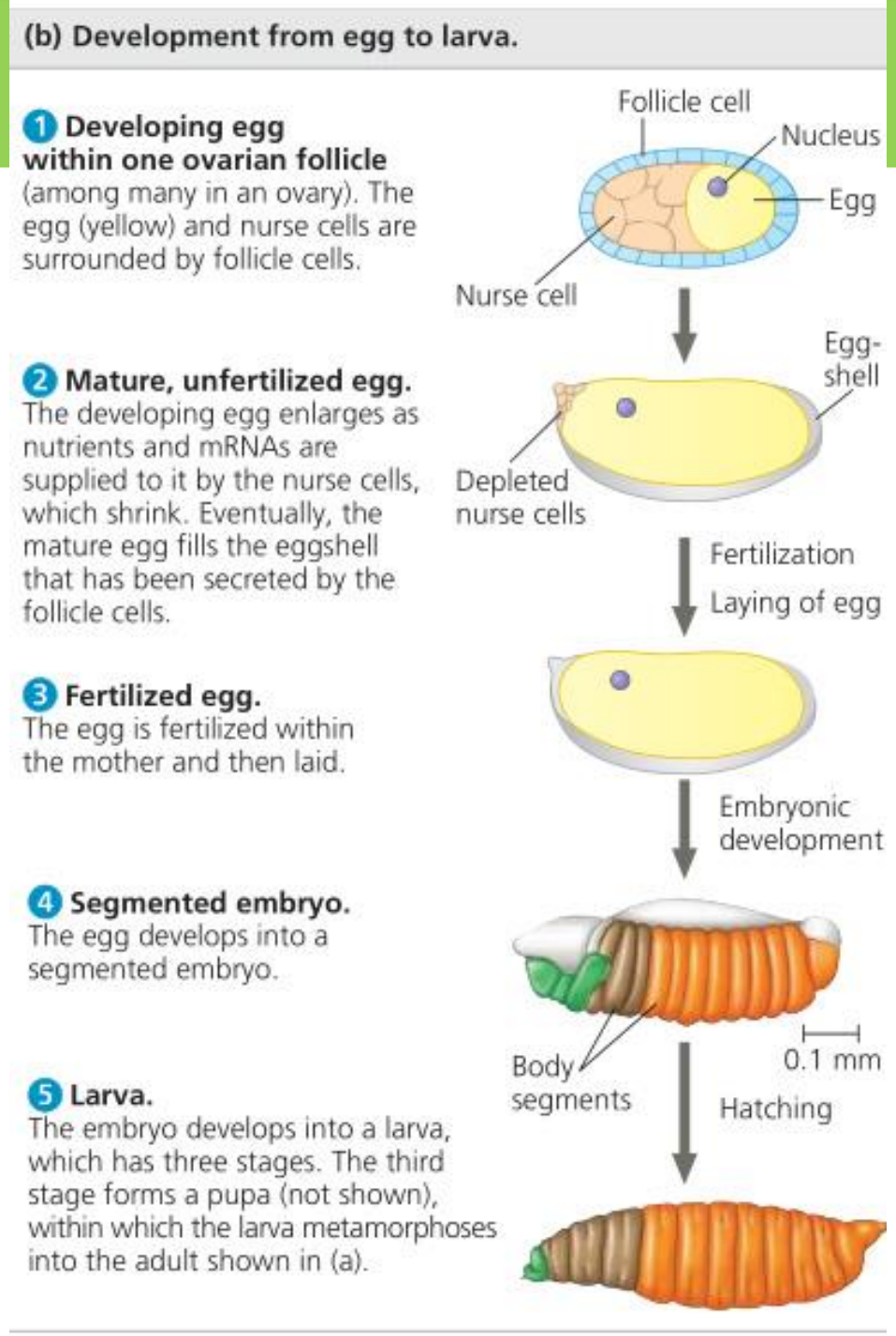
- For the tissues to function effectively in the organism as a whole, the organism's body plan—its over-all three-dimensional arrangement—must be established and superimposed on the differentiation process.
- Pattern formation is the development of a spatial organization of tissues and organs.
- Cytoplasmic determinants and inductive signals both contribute to pattern formation.
- In animals, pattern formation begins with the establishment of the major axes.



- Positional information, the molecular cues (cytoplasmic determinants and inductive signals) that control pattern formation, tells a cell its location relative to the body axes and to neighboring cells.

The Life Cycle of Drosophila

- In Drosophila, cytoplasmic determinants in the unfertilized egg determine the axes before fertilization.
- After fertilization, the embryo develops into a segmented larva with three larval stages.

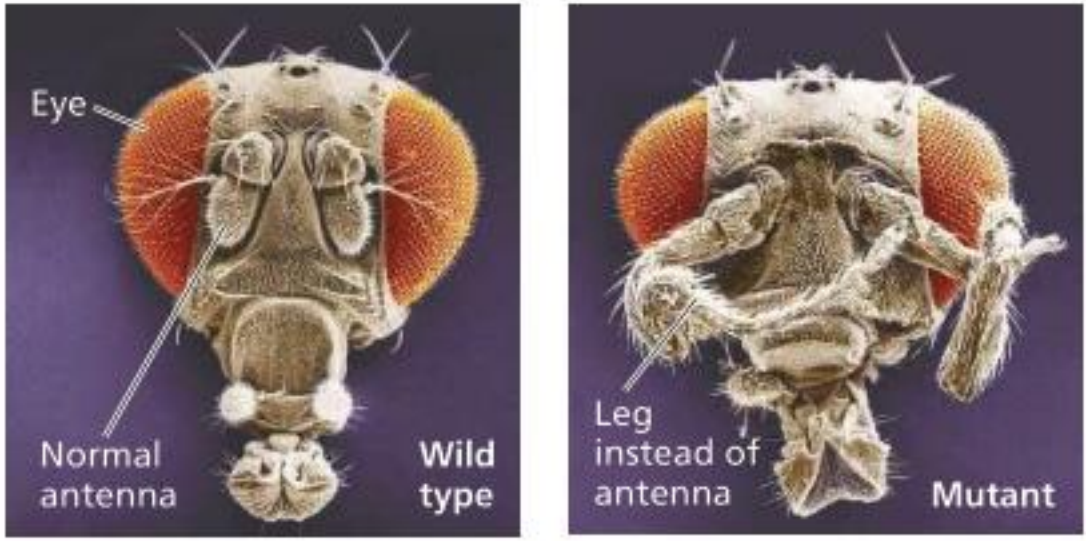


Genetic Analysis of Early Development: Scientific Inquiry

- In the 1940s Edward B. Lewis demonstrated that genes direct the developmental process.

- Lewis studied bizarre mutant flies with developmental defects that led to extra wings or legs in the wrong place

▼ Figure 18.20 Abnormal pattern formation in *Drosophila*.
Mutations in homeotic genes cause misplacement of structures in an animal, such as the legs extending from the mutant fly's head in place of antenna (colorized SEMs).



He located the mutations on the fly's genetic map, thus connecting the developmental abnormalities to specific genes.

The genes Lewis discovered, called homeotic genes, are regulatory genes that control pattern formation in the fly.

Axis Establishment

- Nüsslein-Volhard and Wieschaus set out to identify all the genes that affect segment formation in Drosophila.
- They created mutants, conducted breeding experiments, and looked for corresponding genes.
- They found 120 genes essential for normal segmentation.

- **Maternal effect genes** encode for clytopasmic determinants that initially establish the axes of the body of Drosophila.

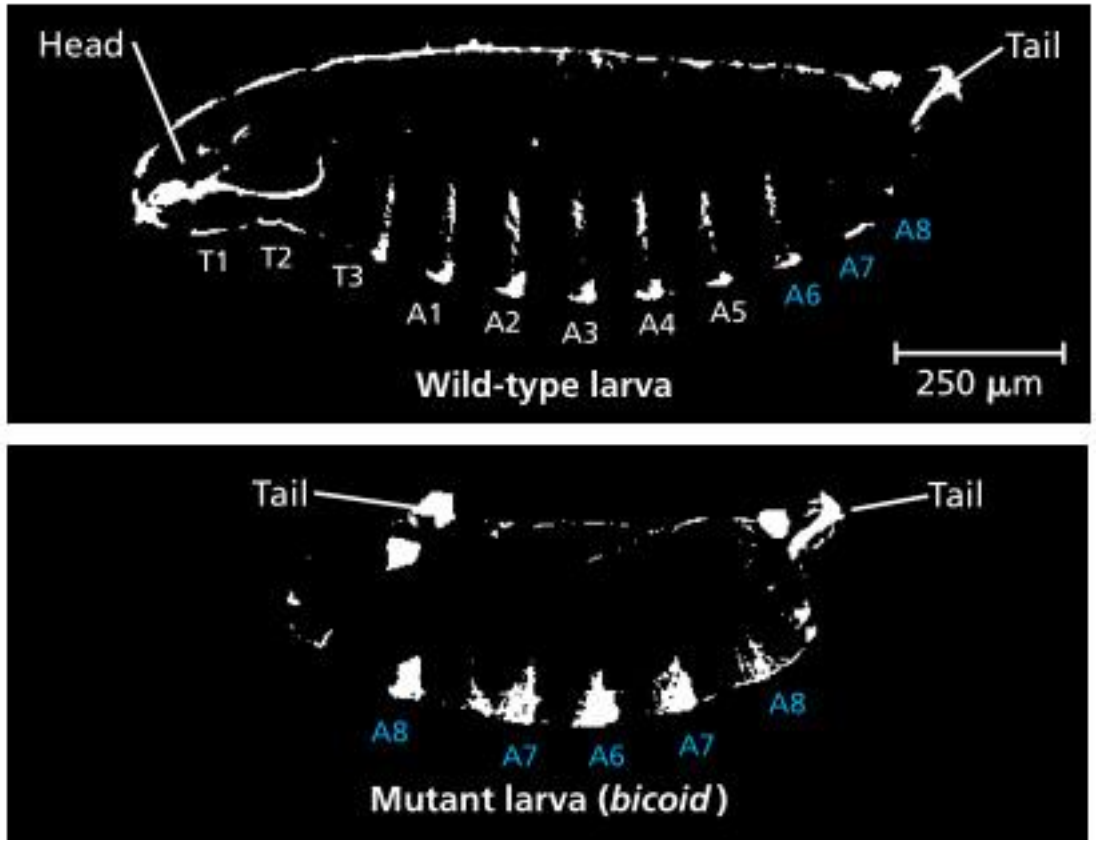
- A a maternal effect gene is one that, when mutant in the mother, results in a mutant phenotype in the offspring, regardless of the offspring's own genotype.

- These maternal effect genes are also called egg-polarity genes because they control orientation of the egg and consequently the fly.

Bicoid: A Morphogen That Determines Head Structures

- One maternal effect gene, the bicoid (two-tailed) gene, affects the front half of the body.
- An embryo whose mother has a mutant bicoid gene lacks the front half of its body and has duplicate posterior structures at both ends.
- This phenotype suggests that the product of the mother's bicoid gene is concentrated at the future anterior end.
- This hypothesis is an example of the morphogen gradient hypothesis, in which gradients of substances called morphogens establish an embryo's axes and other features of its form.

▼ Figure 18.21 Effect of the *bicoid* gene on *Drosophila* development. A wild-type fruit fly larva has a head, three thoracic (T) segments, eight abdominal (A) segments, and a tail. A larva whose mother has two mutant alleles of the *bicoid* gene has two tails and lacks all anterior structures (LMs).

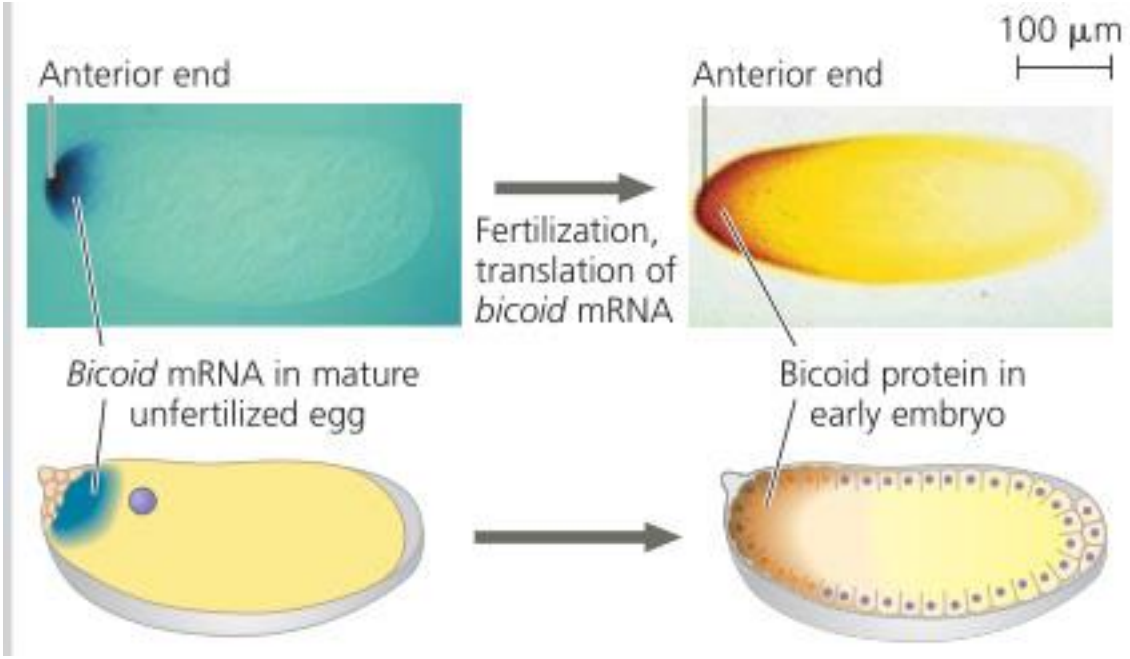


Bicoid: A Morphogen That Determines Head Structures

- Hypothesis: Bicoid gene normally codes for a morphogen that specifies the head (anterior) end of the embryo.
- Experiment: Using molecular techniques Christiane Nüsslein-Volhard and colleagues analyzed expression of the bicoid gene to show whether the mRNA and protein encoded by this gene were found in the anterior end of the fertilized egg and early embryo of wild-type flies.
- Results: Bicoid mRNA (dark blue in the light micrographs and drawings) was confined to the anterior end of the unfertilized egg.

Later in development, Bicoid protein (dark orange) was seen to be concentrated in cells at the anterior end of the embryo.

- Conclusion The location of bicoid mRNA and the diffuse gradient of Bicoid protein seen later are consistent with the hypothesis that Bicoid protein is a morphogen specifying formation of head-specific structures.



Bicoid: A Morphogen That Determines Head Structures

The bicoid research is important for three reasons:

- It identified a specific protein required for some early steps in pattern formation
- It increased understanding of the mother's role in embryo development
- It demonstrated a key developmental principle that a gradient of molecules can determine polarity and position in the embryo

Cancer results from genetic changes that affect cell cycle control

Cancer is a disease in which cells escape from their normal cell division control mechanisms.

The gene regulation systems that go wrong during cancer are the same systems that play important roles in embryonic development, the immune response, and many other biological processes.

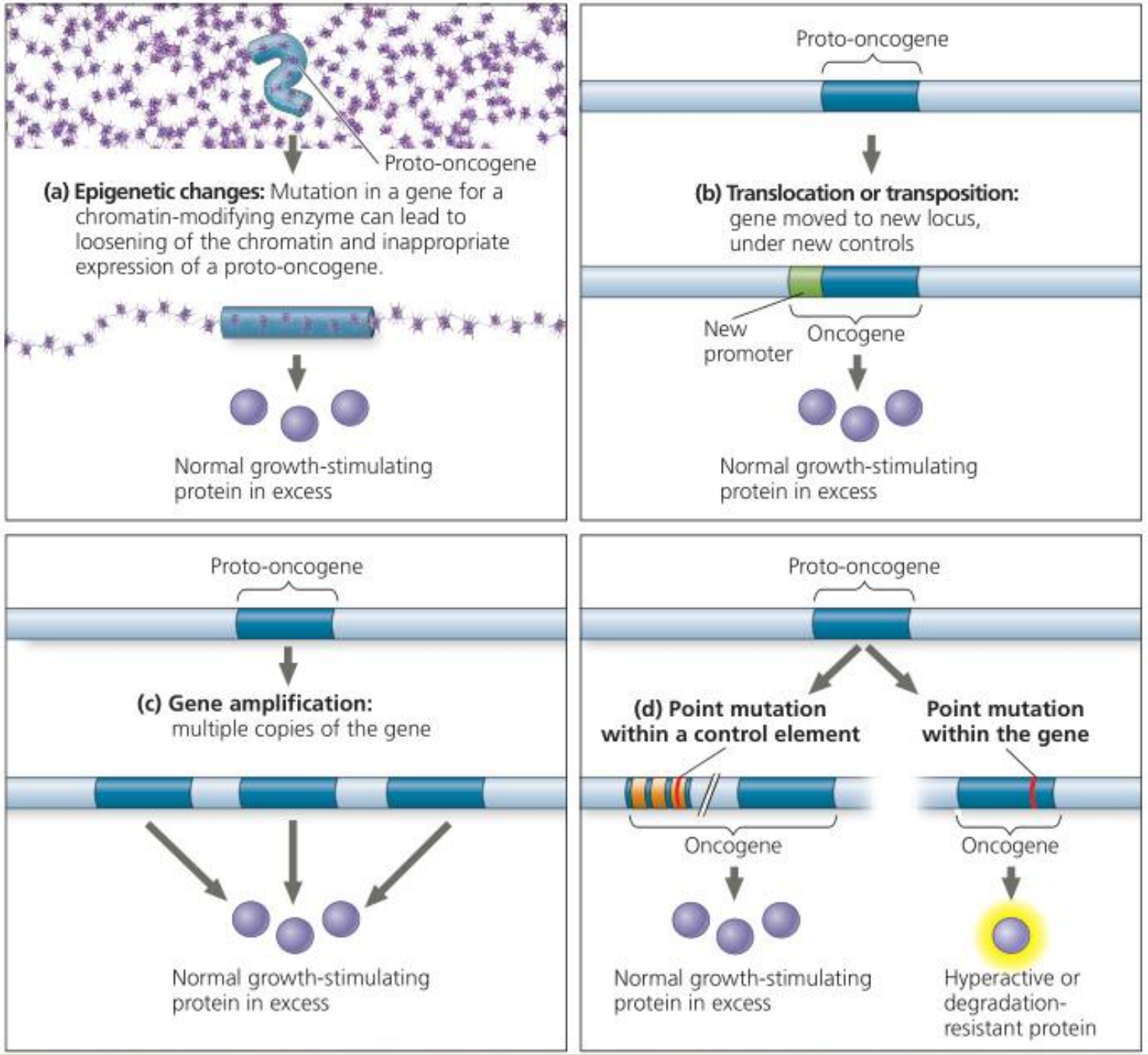
Types of Genes Associated with Cancer

- The genes that regulate cell growth and division are the genes for growth factors, their receptors, and the intracellular molecules of signaling pathways.
- Mutations that alter any of these genes in somatic cells can lead to cancer.
- These can be random spontaneous mutations.
- Many cancer-causing mutations likely also result from environmental influences: chemical carcinogens like tobacco, X-rays and other high-energy radiation, and some viruses.
- Tumor viruses can cause cancer in animals including humans.

Types of Genes Associated with Cancer

- **Oncogenes** are cancer-causing genes.
- **Proto-oncogenes** are the corresponding normal cellular genes that are responsible for normal cell growth and division.
- Conversion of a proto-oncogene to an oncogene can lead to abnormal stimulation of the cell cycle.
- The genetic changes that convert proto-oncogenes to oncogenes fall into four main categories: epigenetic changes, translocations, gene amplification, and point mutations.

▼ **Figure 18.23** **Genetic changes that can turn proto-oncogenes into oncogenes.**



a) If a mutation in a gene for a chromatin-modifying enzyme leads to loosening of chromatin in a region that is normally not being expressed, a proto-oncogene in that region could be expressed at abnormally high levels

b) Cancer cells are frequently found to contain chromosomes that have broken and rejoined incorrectly, translocating fragments from one chromosome to another. If a translocated proto-oncogene ends up near an especially active promoter (or other control element), its transcription may increase, making it an oncogene.

c) amplification, increases the number of copies of the proto-oncogene in the cell through repeated gene duplication.

d) A point mutation either in the promoter or an enhancer that controls a proto-oncogene, could cause an increase in its expression. A point mutation in the coding sequence of the proto-oncogene could change the gene's product to a protein that is more active or more resistant to degradation than the normal protein.

Tumor-Suppressor Genes

- **Tumor-suppressor genes** whose protein products inhibit cell division help prevent uncontrolled cell growth.
- Any mutation that decreases the normal activity of a tumor-suppressor protein may contribute to the onset of cancer, in effect stimulating growth through the absence of suppression.

- Functions of Tumor-suppressor proteins
  - Repair damaged DNA which prevents the cell from accumulating cancer-causing mutations
  - Control cell adhesion. Proper cell anchorage is crucial in normal tissues and is often absent in cancers.
  - Inhibit the cell cycle in the cell-signaling pathway

Interference with Normal Cell-Signaling Pathways

- Mutations in the ras proto-oncogene (30%) and p53 tumor-suppressor gene (50%) are common in human cancers.

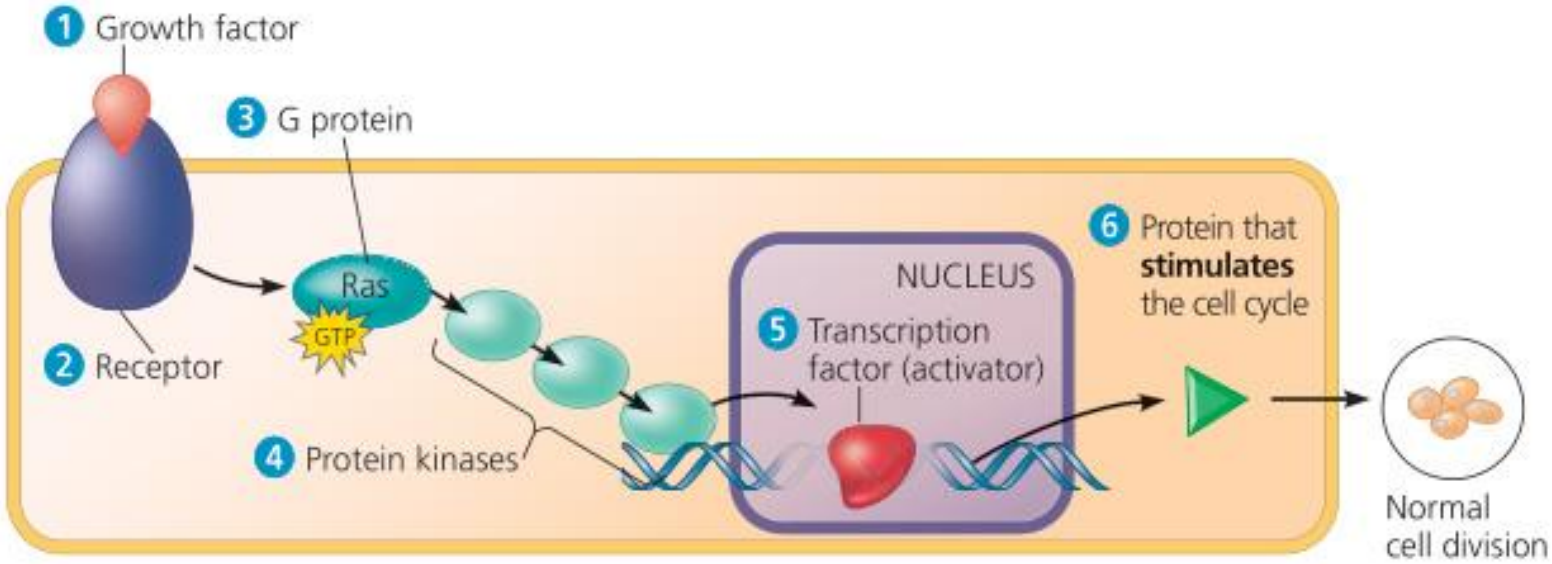
- Mutations in the ras gene can lead to production of a hyperactive Ras protein and increased cell division.

- Suppression of the cell cycle can be important in the case of damage to a cell's DNA; p53 prevents a cell from passing on mutations due to DNA damage.
- The protein p53 encodes is a specific transcription factor that promotes the synthesis of cell cycle–inhibiting proteins.
- Mutations in the p53 gene prevent suppression of the cell cycle and lead to excessive cell growth and cancer.

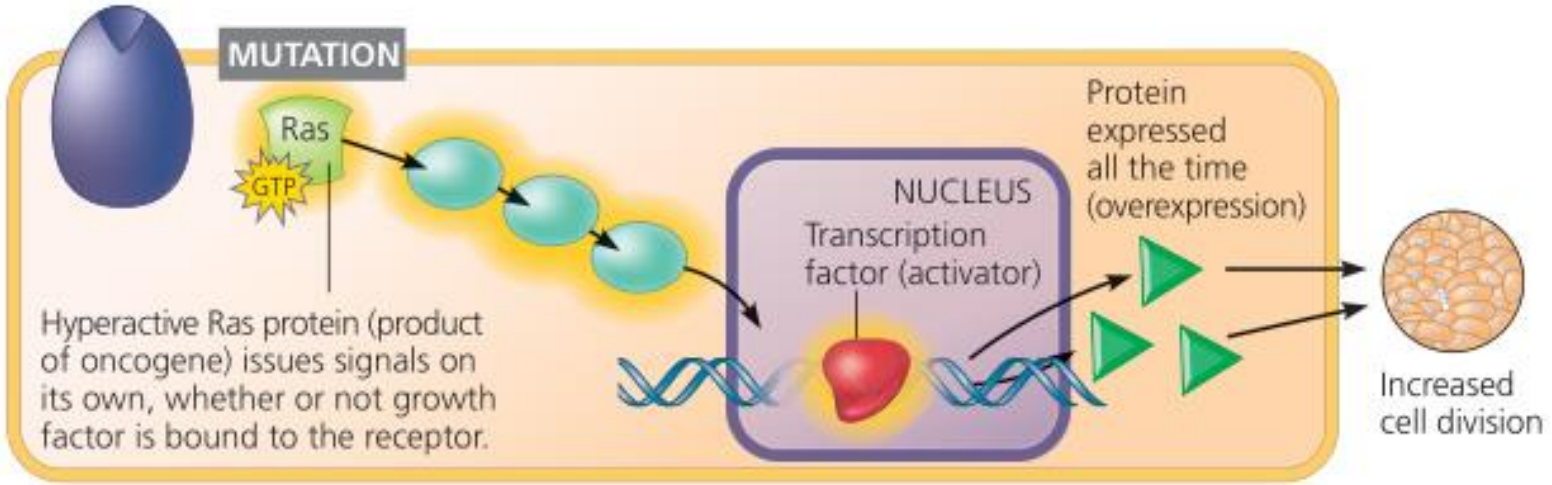
The **Ras protein**, encoded by the ras gene is a G protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of proteins. The cellular response at the end of the pathway is the synthesis of a protein that stimulates the cell cycle.

▼ Figure 18.24 Normal and mutant cell cycle–stimulating pathway.

(a) Normal cell cycle–stimulating pathway. The normal pathway is triggered by 1 a growth factor that binds to 2 its receptor in the plasma membrane. The signal is relayed to 3 a G protein called Ras. Like all G proteins, Ras is active when GTP is bound to it. Ras passes the signal to 4 a series of protein kinases. The last kinase activates 5 a transcription factor (activator) that turns on one or more genes for 6 a protein that stimulates the cell cycle.



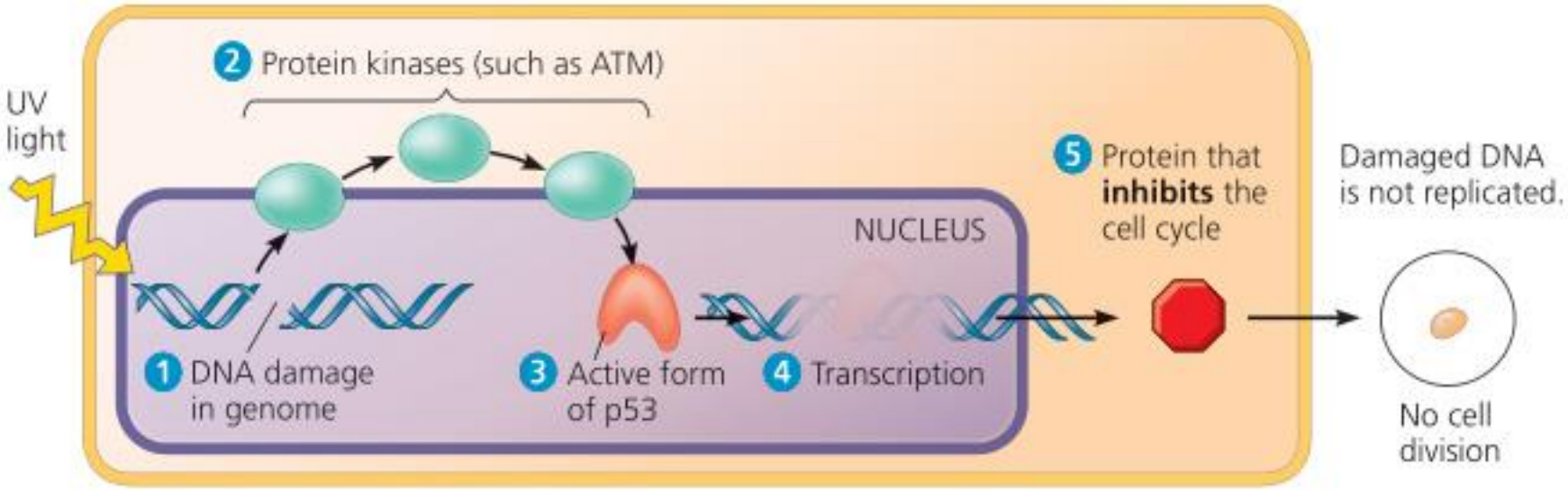
(b) Mutant cell cycle–stimulating pathway. If a mutation makes Ras or any other pathway component abnormally active, excessive cell division and cancer may result.



Cancer among elephants has been estimated at about 3%, compared with 30% for humans. Genome sequencing revealed that elephants have 20 copies of the **p53 gene**, compared to one copy in humans.

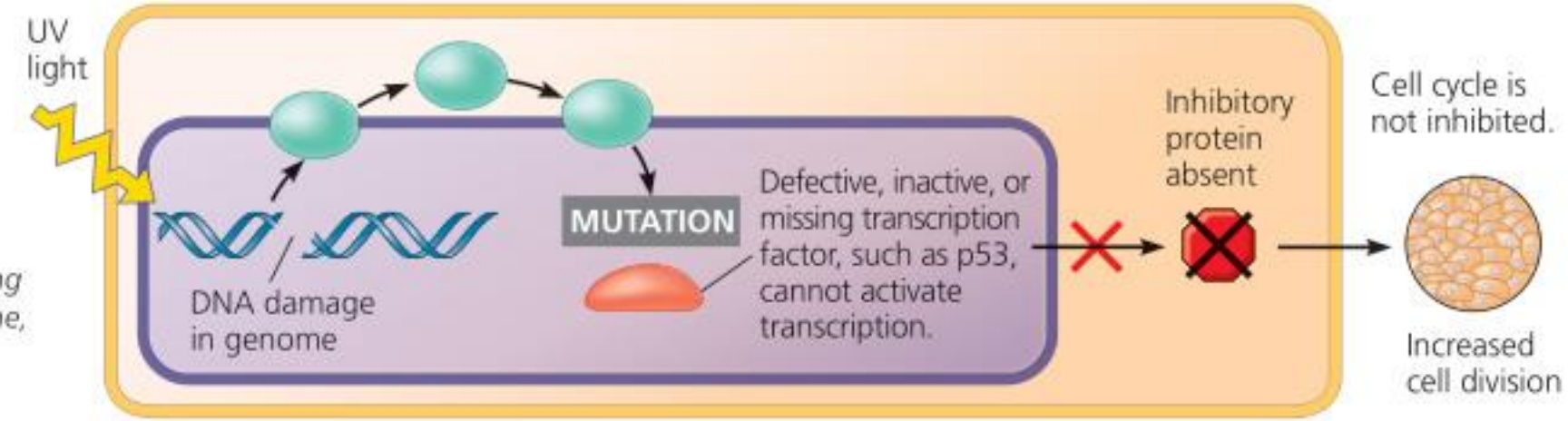
▼ Figure 18.25 Normal and mutant cell cycle–inhibiting pathway.

(a) Normal cell cycle–inhibiting pathway. In the normal pathway, 1 DNA damage is an intracellular signal that is passed via 2 protein kinases, leading to activation of 3 p53. 4 Activated p53 promotes transcription of the gene for 5 a protein that inhibits the cell cycle. The resulting suppression of cell division ensures that the damaged DNA is not replicated. If the DNA damage is irreparable, then the p53 signal leads to apoptosis.



(b) Mutant cell cycle–inhibiting pathway. Mutations causing deficiencies in any pathway component can contribute to the development of cancer.

? *Explain whether a cancer-causing mutation in a tumor-suppressor gene, such as p53, is more likely to be a recessive or a dominant mutation.*



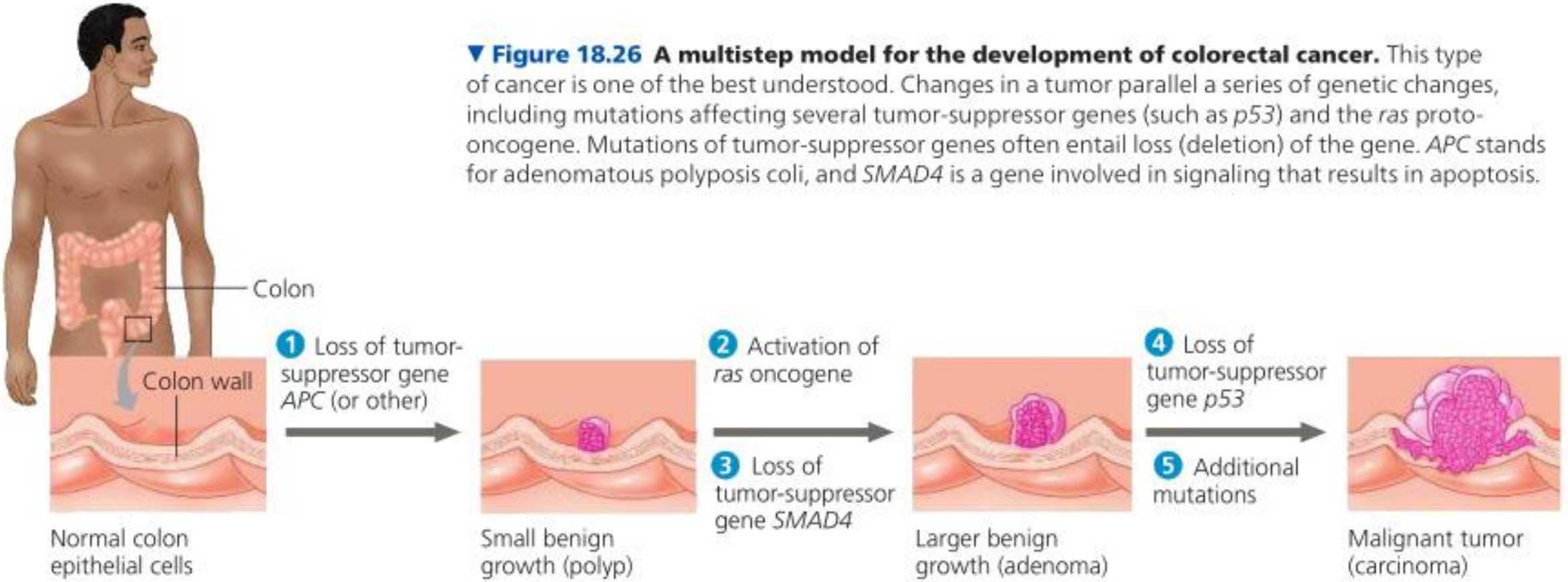
The Multistep Model of Cancer Development

- Multiple mutations are generally needed for full-fledged cancer; thus the incidence increases with age.

- At the DNA level, a cancerous cell is usually characterized by at least one active oncogene and the mutation of several tumor-suppressor genes.

- Like most cancers, colorectal cancer develops gradually
  - The first sign is often a polyp,a small, benign growth in the colon lining.
  - The tumor grows and may eventually become malignant, invading other tissues.
  - The development of a malignant tumor is paralleled by a gradual accumulation of mutations that convert proto-oncogenes to oncogenes and knock out tumor-suppressor genes. A ras oncogene and a mutated p53 tumor-suppressor gene are often involved.
  - Since we understand the progression of this type of cancer, routine screenings (colonoscopies, for example) are recommended to identify and remove any suspicious polyps.

The Multistep Model of Cancer Development



▼ **Figure 18.26 A multistep model for the development of colorectal cancer.** This type of cancer is one of the best understood. Changes in a tumor parallel a series of genetic changes, including mutations affecting several tumor-suppressor genes (such as *p53*) and the *ras* proto-oncogene. Mutations of tumor-suppressor genes often entail loss (deletion) of the gene. *APC* stands for adenomatous polyposis coli, and *SMAD4* is a gene involved in signaling that results in apoptosis.

Inherited Predisposition and Other Factors Contributing to Cancer

- The fact that multiple genetic changes are required to produce a cancer cell helps explain the observation that cancers can run in families.
- An individual inheriting an oncogene or a mutant allele of a tumor-suppressor gene is one step closer to accumulating the necessary mutations for cancer to develop than is an individual without any such mutations.

- Individuals can inherit oncogenes or mutant alleles of tumor-suppressor genes.

- Inherited mutations in the tumor-suppressor gene adenomatous polyposis coli (APC) are common in individuals with colorectal cancer.

- Mutations in the BRCA1 or BRCA2 gene are found in at least half of inherited breast cancers.

The Role of Viruses in Cancer

- A number of tumor viruses can cause cancer in various animals, including humans.
- Viruses play a role in about 15% of the cases of human cancer.
- viruses can interfere with gene regulation in several ways if they integrate their genetic material into the DNA of a cell.
- Viral integration may donate an oncogene to the cell, disrupt a tumor-suppressor gene, or convert a proto-oncogene to an oncogene.
- Some viruses produce proteins that inactivate p53 and other tumor-suppressor proteins, making the cell more prone to becoming cancerous.